

Creating a Wi-Fi honeypot

As an aspiring penetration tester, you may be asked to conduct extensive wireless security testing for your company or a client organization. Creating a rogue access point with an interesting SSID (wireless network name), such as `VIP_WiFi` or `Company-name_VIP`, will lure employees to establish a connection.

When creating a rogue access point, the objective is to capture user credentials and sensitive information, as well as detecting any vulnerable wireless clients in an organization. The following are some tips to consider when deploying your rogue access point:

- Choose a suitable location to ensure there is maximum coverage for potential victims.
- De-authenticate clients from the real access point, causing them to create an association with the rogue access point.
- Create a captive portal to capture user credentials.

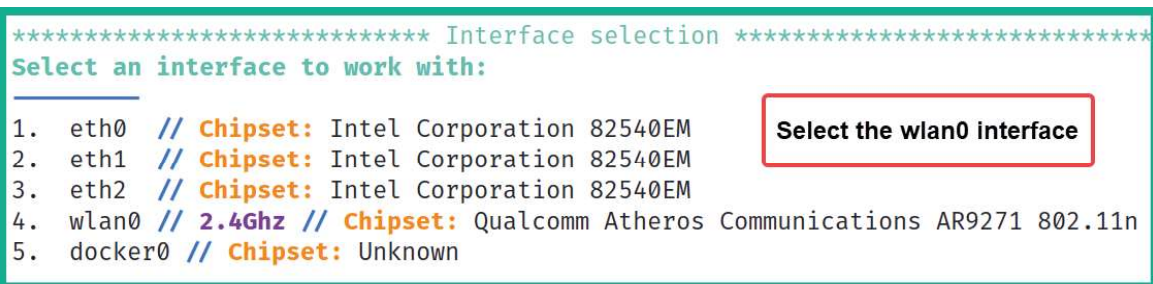
To get started, we are going to use Airgeddon once more as it contains a lot of features and functions that will assist us with gathering information about a target wireless network and its clients. It will also help us launch various types of attacks and lure users to associate with our rogue access point.

To get started with this exercise, please follow these steps:

1. Power on Kali Linux and ensure it has an internet connection and that a wireless network adapter is connected.
2. Next, open Terminal and use the following command to start Airgeddon:

```
kali@kali:~$ sudo airgeddon
```

3. Next, select your wireless network adapter to perform the attack:



```
***** Interface selection *****
Select an interface to work with:

1. eth0 // Chipset: Intel Corporation 82540EM
2. eth1 // Chipset: Intel Corporation 82540EM
3. eth2 // Chipset: Intel Corporation 82540EM
4. wlan0 // 2.4Ghz // Chipset: Qualcomm Atheros Communications AR9271 802.11n
5. docker0 // Chipset: Unknown
```

Figure 13.52 – Selecting a wireless network adapter

4. Next, enable monitor mode on your wireless adapter by selecting option 3:

```
***** airgeddon v10.42 main menu *****
Interface wlan0 selected. Mode: Managed. Supported bands: 2.4Ghz

Select an option from menu:
0. Exit script
1. Select another network interface
2. Put interface in monitor mode
3. Put interface in managed mode
```



Figure 13.53 – Enabling monitor mode

5. Next, select option 7 to access **Evil Twin attacks menu**:

```
***** airgeddon v10.42 main menu *****
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz

Select an option from menu:
0. Exit script
1. Select another network interface
2. Put interface in monitor mode
3. Put interface in managed mode
4. DoS attacks menu
5. Handshake/PMKID tools menu
6. Offline WPA/WPA2 decrypt menu
7. Evil Twin attacks menu
8. WPS attacks menu
9. WEP attacks menu
10. Enterprise attacks menu
```



Figure 13.54 – Accessing Evil Twin attacks menu

6. Next, select option 4 to **Explore for targets**:

```

***** Evil Twin attacks menu *****
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz
Selected BSSID: None
Selected channel: None
Selected ESSID: None

Select an option from menu:

0. Return to main menu
1. Select another network interface
2. Put interface in monitor mode
3. Put interface in managed mode
4. Explore for targets (monitor mode needed)
   (without sniffing, just AP)
5. Evil Twin attack just AP

```

Figure 13.55 – Explore for targets

A new window will appear that shows the live scan for nearby access points. In this exercise, the target is Corp_Wi-Fi. Once the target has been found, press *Ctrl* + *C* in the pop-up window to stop the scan and continue:

Exploring for targets										
CH 3][Elapsed: 18 s][2021-09-21 09:24										
BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID	
2C:9D:1E:...	-91	2	0 0	10	195	WPA2	CCMP	PSK	Digicel_WiFi_fh4w	
68:7F:74:01:28:E1	-55	31	0 0	6	130	WPA2	CCMP	PSK	Corp_Wi-Fi	
9C:3D:CF:...	-20	74	0 0	8	540	WPA2	CCMP	PSK	!!><!!	
38:4C:4F:...	-75	20	0 0	1	195	WPA2	CCMP	PSK	Digicel_WiFi_T28R	
BSSID	STATION		PWR	Rate	Lost	Frames	Notes	Probes		
(not associated)	E0:D4:64:...		-17	0 - 1	0	3		!!><!!		
68:7F:74:01:28:E1	D8:50:E6:2F:F9:2B		-36	0 - 6	0	1				

Figure 13.56 – Nearby wireless networks

7. Next, the **Select target** menu will appear. Select the target network and hit *Enter* to continue:

***** Select target *****					
N.	BSSID	CHANNEL	PWR	ENC	ESSID
1)	68:7F:74:01:28:E1	6	62%	WPA2	Corp_Wi-Fi
2)	2C:9D:1E:...	10	11%	WPA2	Digicel_WiFi_fh4w
3)	38:4C:4F:...	1	21%	WPA2	Digicel_WiFi_T28R
4)	9C:3D:CF:...	8	45%	WPA2	!><!

Figure 13.57 – Choosing a target

8. Next, select option 5 to use **Evil Twin attack just AP:**

```
***** Evil Twin attacks menu *****
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz
Selected BSSID: 68:7F:74:01:28:E1
Selected channel: 6
Selected ESSID: Corp_Wi-Fi

Select an option from menu:
_____
0. Return to main menu
1. Select another network interface
2. Put interface in monitor mode
3. Put interface in managed mode
4. Explore for targets (monitor mode needed)
_____ (without sniffing, just AP) _____
5. Evil Twin attack just AP
_____ (with sniffing) _____
6. Evil Twin AP attack with sniffing
7. Evil Twin AP attack with sniffing and bettercap-sslstrip2
8. Evil Twin AP attack with sniffing and bettercap-sslstrip2/BeEF
_____ (without sniffing, captive portal) _____
9. Evil Twin AP attack with captive portal (monitor mode needed)
```

Figure 13.58 – Choosing an attack type

9. Next, select option 2 to perform a de-authentication attack using `aireplay-ng` on clients that are associated with the target wireless network:

```
***** Evil Twin deauth *****
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz
Selected BSSID: 68:7F:74:01:28:E1
Selected channel: 6
Selected ESSID: Corp_Wi-Fi
Selected internet interface: None

Select an option from menu:
_____
0. Return to Evil Twin attacks menu
_____
1. Deauth / disassoc amok mdk4 attack
2. Deauth aireplay attack
3. WIDS / WIPS / WDS Confusion attack
```

Figure 13.59 – Selecting Deauth aireplay attack

10. You will be prompt with the question *Do you want to enable "DoS pursuit mode"?* Type `N` for no and hit *Enter* to continue.

11. Next, select the interface that has an active internet connection on Kali Linux:

```
***** Evil Twin attack just AP *
Select another interface with internet access:

0. Return to Evil Twin attacks menu

1. eth0 // Chipset: Intel Corporation 82540EM
2. eth1 // Chipset: Intel Corporation 82540EM
3. eth2 // Chipset: Intel Corporation 82540EM
4. docker0 // Chipset: Unknown
```

Figure 13.60 – Selecting an internet interface

12. You will be prompted with the question *Do you want to continue?* Type Y for yes and hit *Enter* to continue.

13. Another prompt will ask you, *Do you want to spoof your MAC address during this attack?* Type N for no and hit *Enter* to continue.

Airgeddon will create the following four windows. Each window provides the status of the honeypot, the DHCP service, the de-authentication attack, and an indication of the clients connecting to the honeypot:

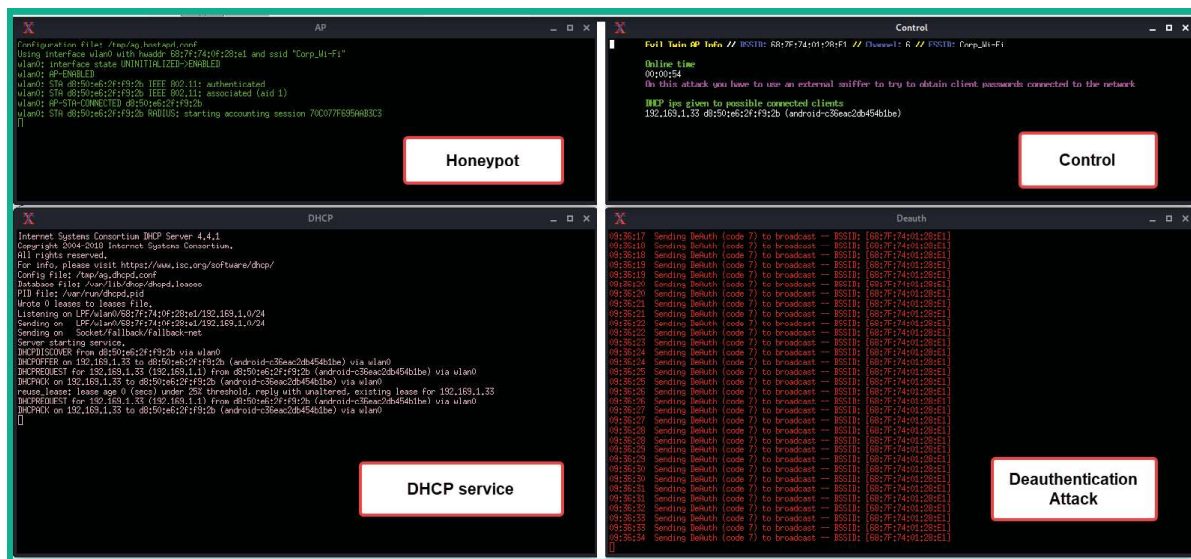


Figure 13.61 – Honeypot in effect

Having completed this section, you have learned how to set up a wireless honeypot using Kali Linux. In the next section, you will learn about WPA3 wireless attacks.